

SmartHire AI

Product Requirements Document

Version: 1.0

Project Type: AI-Powered Recruitment Platform

Prepared By: Tarunteja Patro

Date: July 2026

Table of Contents

1. Executive Summary
2. Problem Statement
3. Proposed Solution
4. Project Scope
5. Stakeholders
6. Target Audience & User Personas
7. User Stories
8. Functional Requirements
9. Non-Functional Requirements
10. Business Rules
11. Acceptance Criteria
12. System Architecture
13. AWS Architecture
14. Database Design
15. API Specifications
16. UI/UX Requirements
17. AI Integration
18. Security
19. Testing Strategy
20. Deployment Strategy
21. Future Roadmap

Executive Summary

SmartHire AI is an AI-powered recruitment platform that assists job seekers in improving their resumes before applying for jobs. Candidates upload resumes in PDF or DOCX format, the system extracts content, and AI evaluates the resume to produce:

- ATS Compatibility Score
- Resume Summary
- Missing Skills
- Resume Strengths & Weaknesses
- Improvement Suggestions
- AI-Generated Interview Questions
- Interview Readiness Score

All analyses are stored in MySQL. Recruiters access a lightweight dashboard to review analyzed candidates and view AI-generated reports.

The application demonstrates Full Stack Development, AI Integration, Cloud Computing (AWS), Database Design, REST APIs, and Responsive UI/UX Design.

Problem Statement

Candidate Challenges

- **Uncertain Quality Assessment:** No objective way to determine if resume meets industry standards
- **Skill Gap Blindness:** Unaware of skills needed for target positions
- **Inadequate Interview Preparation:** Lack of role-specific question practice
- **ATS Complexity:** 85% of resumes fail initial ATS screening

Recruiter Challenges

- **Screening Overwhelm:** 250+ applications per corporate job posting
- **Inconsistent Evaluation:** Manual review varies across recruiters
- **Limited Insights:** No comprehensive candidate analysis beyond basic matching
- **Time-to-Hire Delays:** Top talent typically off market within 10 days

Proposed Solution

SmartHire AI delivers a comprehensive platform with:

Candidate Features:

- Resume upload (PDF, DOCX) and manual text input
- Resume parsing and data extraction
- ATS compatibility scoring
- Comprehensive analysis (summary, skills, strengths, weaknesses)
- Improvement suggestions
- AI-generated interview questions
- Interview readiness score
- Downloadable analysis report

Recruiter Features:

- Candidate list with key metrics

Project Scope

In Scope

Candidate Features:

- Landing page
- Resume upload (PDF, DOCX)
- Manual resume text input
- Resume parsing and content extraction
- ATS compatibility score
- Resume summary
- Keyword analysis
- Skills extraction
- Experience, Education, Projects, Achievements analysis
- Improvement suggestions
- Interview question generation
- Interview readiness score
- Download analysis report

Recruiter Features:

- Candidate list view
- Candidate details view
- ATS score, summary, suggestions, interview score display
- Complete analysis report viewing

Technical Implementation:

- React.js frontend with Tailwind CSS
- Node.js/Express.js backend with REST APIs
- MySQL database
- AWS EC2 and RDS deployment
- Google Gemini API integration

- Responsive UI/UX

Out of Scope

- Authentication (login, signup, password management)
- Admin dashboard
- Applicant Tracking System
- Job posting
- Candidate search/filtering
- Email notifications
- Payments
- Analytics dashboards

Stakeholders

Stakeholder	Role	Key Interests
Developer	Builds and implements solution	Technical implementation, code quality
Project Manager	Oversees development	Timeline, scope, deliverables
End Users (Candidates)	Use the platform	Value, ease of use, accurate results
End Users (Recruiters)	Evaluate candidates	Efficiency, insight quality

Target Audience & User Personas

Candidate: Student Graduate

- **Name:** Priya Sharma, 22, Computer Science graduate
- **Goals:** Optimize resume for technology positions

- **Pain Points:** Limited experience, unsure of ATS requirements
- **Motivation:** Secure first professional role

Candidate: Experienced Professional

- **Name:** Rahul Verma, 32, Software Engineer
- **Goals:** Advance career, improve interview performance
- **Pain Points:** Resume optimization for senior positions
- **Motivation:** Career advancement and compensation

Recruiter

- **Name:** Ananya Reddy, 29, Technical Recruiter
- **Goals:** Efficient candidate screening
- **Pain Points:** Time spent reviewing resumes manually
- **Motivation:** Quality hires, reduced time-to-fill

User Stories

ID	User Story	Priority
US-001	As a job seeker, I want to upload my resume (PDF/DOCX)	High
US-002	As a job seeker, I want to manually input resume text	Medium
US-003	As a job seeker, I want an ATS compatibility score	High
US-004	As a job seeker, I want a resume summary	High
US-005	As a job seeker, I want to see missing skills	High
US-006	As a job seeker, I want to understand resume strengths/weaknesses	High

US-007	As a job seeker, I want improvement suggestions	High
US-008	As a job seeker, I want AI-generated interview questions	High
US-009	As a job seeker, I want an interview readiness score	High
US-010	As a job seeker, I want to download my analysis report	High
US-011	As a recruiter, I want to view candidate list and details	High
US-012	As a recruiter, I want to view ATS scores and interview scores	High

Functional Requirements

ID	Requirement	Description
FR-001	Landing Page	Product name, value proposition, features, CTA
FR-002	File Upload	Accept PDF (max 10MB) and DOCX (max 10MB)
FR-003	Manual Input	Text area input (max 50,000 characters)
FR-004	Resume Parsing	Extract: name, email, phone, experience, education, skills, projects, achievements
FR-005	ATS Score	Generate 0-100 score with breakdown and recommendations
FR-006	Resume Summary	Professional summary with key highlights
FR-007	Keyword Analysis	Important keywords, density analysis, recommendations

FR-008	Skills Extraction	Technical/soft skills, proficiency levels, missing skills with priority
FR-009	Experience Analysis	Years, role depth, relevance assessment
FR-010	Education Analysis	Degrees, institutions, relevance
FR-011	Projects Analysis	Descriptions, technologies, outcomes
FR-012	Achievements Analysis	Quantifiable achievements, impact
FR-013	Improvement Suggestions	Specific, actionable, prioritized recommendations
FR-014	Interview Questions	Technical, behavioral, situational, role-specific
FR-015	Readiness Score	0-100 score with explanation and improvement areas
FR-016	Report Download	Comprehensive, professional, print-friendly report
FR-017	Recruiter Dashboard	Candidate list with key metrics, sorting
FR-018	Candidate Details	Complete analysis, all sections, comprehensive view

ID	Requirement	Target
NFR-001	Performance	Page load < 3s, API response < 2s, AI analysis < 30s
NFR-002	Reliability	99.5% uptime, graceful error handling, data persistence
NFR-003	Security	File validation, input sanitization, SQL injection prevention, rate limiting
NFR-004	Usability	Responsive design, intuitive navigation, clear error messages
NFR-005	Scalability	Support concurrent users, connection pooling, horizontal scaling readiness
NFR-006	Maintainability	Modular architecture,

comprehensive documentation, version control		
ID	Rule	Description
BR-001	File Type Validation	Only PDF and DOCX accepted
BR-002	File Size Limit	Maximum 10MB
BR-003	Manual Text Limit	Maximum 50,000 characters
BR-004	Score Range	ATS and Interview scores: 0-100
BR-005	Data Preservation	All analysis data stored permanently
BR-006	Report Generation	Only for completed analyses

BR-007	Recruiter Access	Can view all candidate analyses
Acceptance Criteria		
ID	Feature	Acceptance Criteria
AC-001	Resume Upload	PDF/DOCX accepted; invalid files rejected with error
AC-002	Resume Parsing	All sections correctly extracted
AC-003	ATS Score	0-100 score with explanation and breakdown
AC-004	Resume Summary	Accurately reflects content, professionally formatted
AC-005	Missing Skills	Identified, relevant, prioritized list

AC-006	Strengths/Weaknesses	Identified with evidence, balanced feedback
AC-007	Improvement Suggestions	Actionable, specific, prioritized, with guidance
AC-008	Interview Questions	Relevant, multiple types, role-appropriate
AC-009	Readiness Score	0-100 score with explanation and recommendations
AC-010	Report Download	Complete sections, professional formatting, printable
AC-011	Recruiter Dashboard	All candidates displayed, key metrics visible, intuitive
AC-012	Candidate Details	Complete analysis, all sections, comprehensive view

Non-Functional Requirements

System Architecture

SmartHire AI follows a modern three-tier architecture:

Architectural Overview

Presentation Layer: React.js single-page application

Application Layer: Node.js/Express.js REST API

Data Layer: MySQL database

AI Layer: Google Gemini API integration

Cloud Layer: AWS infrastructure

Component Interaction

Upload Flow: User submits file → Frontend sends to backend → Backend validates → Extracts content → Sends to Gemini API → AI processes → Results stored in MySQL → Results returned to frontend

Recruiter Flow: Recruiter accesses dashboard → Frontend requests data → Backend queries database → Returns data → Frontend displays

Frontend Architecture

- **Framework:** React.js
- **Styling:** Tailwind CSS
- **HTTP Client:** Axios
- **Routing:** React Router

Component Structure:

text

App

- |— LandingPage (Header, Hero, Features, Footer)
- |— UploadPage (FileUpload, ManualInput, UploadButton)
- |— AnalysisPage (ATSScore, Summary, Skills, Strengths,
| Suggestions, Questions, Readiness, Download)

- └─ RecruiterDashboard (CandidateList, CandidateDetails)
- └─ Common (LoadingSpinner, ErrorBoundary, Notification)

Backend Architecture

- **Framework:** Express.js
- **File Upload:** Multer
- **AI Integration:** Gemini API client
- **Database:** MySQL2 driver

Architecture Layers:

text

Backend

- └─ Controllers (Analysis, Resume, Recruiter)
- └─ Services (FileParser, AI, Analysis, Database)
- └─ Models (Candidate, Resume, Analysis, Skill, Experience, Education, Project, Question)
- └─ Middleware (FileValidation, ErrorHandler, RateLimiter, CORS)
- └─ Routes (analysis, resume, recruiter)

AWS Architecture

Infrastructure Components

AWS EC2:

- AMI: Ubuntu 22.04 LTS
- Instance Type: t2.micro (or t2.medium)
- Node.js runtime environment
- Security groups for controlled access

AWS RDS:

- Engine: MySQL 8.0
- Class: db.t2.micro (or db.t3.micro)

- Automated backups (7 days retention)
- Security group restrictions

Network Security:

- VPC for network isolation
- Security groups restrict unnecessary access
- Database accessed only from application server
- Public access limited to HTTP/HTTPS ports

Database Design

ER Diagram Description

Relationships:

- Candidate ↔ Resume: One-to-One
- Resume ↔ Analysis: One-to-One
- Candidate ↔ Skills: One-to-Many
- Candidate ↔ Experiences: One-to-Many
- Candidate ↔ Educations: One-to-Many
- Candidate ↔ Projects: One-to-Many
- Analysis ↔ InterviewQuestions: One-to-Many

Database Tables			
Candidate Table:			
Column	Data Type	Constraints	Description
candidate_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique identifier
full_name	VARCHAR(100)	NOT NULL	Candidate's full name
email	VARCHAR(100)	NOT NULL	Email address
phone	VARCHAR(20)		Phone number
location	VARCHAR(100)		City and state
linkedin_url	VARCHAR(255)		LinkedIn profile
portfolio_url	VARCHAR(255)		Portfolio website
professional_summary	TEXT		Brief summary
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Creation timestamp

Resume Table:

Column	Data Type	Constraints	Description
resume_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique identifier
candidate_id	INT	FOREIGN KEY	References candidate
file_name	VARCHAR(255)		Original file name
file_type	VARCHAR(20)		PDF, DOCX, or TEXT
file_size	INT		File size in bytes
extracted_text	LONGTEXT		Full extracted content
upload_source	VARCHAR(20)		FILE or MANUAL
upload_date	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Upload timestamp

Analysis Table:

Column	Data Type	Constraints	Description
analysis_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique identifier
resume_id	INT	FOREIGN KEY	References resume
ats_score	INT	0-100	ATS compatibility score
ats_breakdown	TEXT		Score breakdown
resume_summary	TEXT		Generated summary
strengths	TEXT		List of strengths
weaknesses	TEXT		List of weaknesses
missing_skills	TEXT		Missing skills list
suggestions	TEXT		Improvement suggestions
interview_readiness_score	INT	0-100	Readiness score

readiness_explanation	TEXT		Score explanation
ai_response_raw	LONGTEXT		Raw AI response
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Analysis timestamp
Skills Table:			

Column	Data Type	Constraints	Description
skill_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique identifier
candidate_id	INT	FOREIGN KEY	References candidate
skill_name	VARCHAR(100)	NOT NULL	Skill name
skill_category	VARCHAR(50)		Technical or Soft
proficiency_level	VARCHAR(20)		Beginner, Intermediate, Expert
is_missing	BOOLEAN	DEFAULT FALSE	Missing skill flag
priority	INT	1-5	Importance priority

Experiences Table:

Column	Data Type	Constraints	Description
experience_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique identifier
candidate_id	INT	FOREIGN KEY	References candidate
job_title	VARCHAR(100)	NOT NULL	Job title
company_name	VARCHAR(100)	NOT NULL	Company name
location	VARCHAR(100)		Work location
start_date	DATE		Start date
end_date	DATE		End date
current_role	BOOLEAN	DEFAULT FALSE	Current role flag
description	TEXT		Job description
achievements	TEXT		Key achievements

Educations Table:

Column	Data Type	Constraints	Description
education_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique identifier
candidate_id	INT	FOREIGN KEY	References candidate
degree	VARCHAR(100)	NOT NULL	Degree obtained
field_of_study	VARCHAR(100)		Field of study
institution	VARCHAR(100)	NOT NULL	Institution name
location	VARCHAR(100)		Institution location
start_date	DATE		Start date
end_date	DATE		End date
gpa	DECIMAL(3,2)		Grade point average
Projects Table:			
Column	Data Type	Constraints	Description
project_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique identifier

candidate_id	INT	FOREIGN KEY	References candidate
project_name	VARCHAR(100)	NOT NULL	Project name
description	TEXT		Project description
technologies	TEXT		Technologies used
outcomes	TEXT		Project outcomes
project_url	VARCHAR(255)		Project URL

InterviewQuestions Table:

Column	Data Type	Constraints	Description
question_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique identifier
analysis_id	INT	FOREIGN KEY	References analysis
question_text	TEXT	NOT NULL	Interview question
question_type	VARCHAR(50)		Technical, Behavioral, Situational

difficulty	VARCHAR(20)	Easy, Medium, Hard
suggested_answer	TEXT	Suggested answer
category	VARCHAR(50)	Question category

API Specifications

Base URL

Development: `http://localhost:5000/api/v1`

Common Response Format

Success:

```
json
{
  "success": true,
  "data": { ... },
  "message": "Operation successful",
  "timestamp": "2026-07-21T10:30:00Z"
}
```

Error:

```
json
{
  "success": false,
  "error": "Error message",
  "code": "ERROR_CODE",
}
```

```
"details": "Additional details",  
"timestamp": "2026-07-21T10:30:00Z"  
}
```

REST Endpoints

Method	Endpoint	Description	Request Body	Response
POST	/analysis/upload-file	Upload PDF/DOCX	Multipart file	Analysis result
POST	/analysis/upload-text	Manual text input	JSON with resume_text	Analysis result
GET	/analysis/:id	Get analysis by ID	URL parameter	Analysis data
GET	/analysis/:id/report	Download report	URL parameter	Report file
GET	/candidates	Get all candidates	None	Candidate list
GET	/candidates/:id	Get candidate by ID	URL parameter	Candidate details
GET	/candidates/:id/analysis	Get candidate analysis	URL parameter	Analysis data

Request Examples

Upload File:

text

POST /api/v1/analysis/upload-file

Content-Type: multipart/form-data

--boundary

Content-Disposition: form-data; name="resume"; filename="resume.pdf"

Content-Type: application/pdf

[Binary file data]

--boundary--

Upload Manual Text:

```
json
{
  "resume_text": "John Doe\nSoftware Engineer...",
  "candidate_name": "John Doe",
  "candidate_email": "john@example.com"
}
```

Response Example

Analysis Success:

```
json
{
  "success": true,
  "data": {
    "analysis_id": 1,
    "ats_score": 78,
    "ats_breakdown": "Keyword density: 75%, Formatting: 85%, Structure: 75%",
    "resume_summary": "Senior software engineer with 5+ years experience...",
    "strengths": ["Strong technical skills", "Proven leadership", "Excellent communication"],
    "weaknesses": ["Limited cloud experience", "Need more quantifiable achievements"],
    "missing_skills": ["AWS", "Kubernetes", "CI/CD"],
    "suggestions": ["Add cloud experience", "Quantify achievements", "Expand technical descriptions"],
    "interview_questions": [
      {"question": "Explain microservices architecture", "type": "Technical", "difficulty": "Hard"}
    ],
    "interview_readiness_score": 72,
    "readiness_explanation": "Good technical foundation, needs deeper system design preparation",
    "message": "Analysis completed successfully",
    "timestamp": "2026-07-21T10:30:00Z"
  }
}
```

Status Codes

Status Code	Description	Usage
200	Success	GET requests
201	Created	Analysis created
400	Bad Request	Invalid input
413	Payload Too Large	File exceeds limit
415	Unsupported Media Type	Invalid file type
500	Internal Server Error	Server/AI failure
503	Service Unavailable	AI/database unavailable

Error Codes

Error Code	Status	Description	Solution
FILE_TYPE_INVALID	400	File not supported	Upload PDF or DOCX
FILE_TOO_LARGE	413	File exceeds limit	Compress file or use manual input
TEXT_TOO_LONG	400	Text exceeds limit	Shorten text
PARSING_FAILED	500	Failed to parse file	Retry or use manual input
AI_SERVICE_ERROR	503	Gemini API error	Retry later
DATABASE_ERROR	500	Database failure	Contact support
ANALYSIS_NOT_FOUND	404	Analysis not found	Verify ID

UI/UX Requirements

Visual Design

Primary Colors:

- Primary Blue: #2563EB
- Primary Dark: #1D4ED8
- Primary Light: #60A5FA

Accent Colors:

- Success: #059669
- Warning: #D97706
- Error: #DC2626

Score Indicators:

- 0-40: Red (#DC2626)
- 41-70: Orange (#D97706)
- 71-100: Green (#059669)

Typography

- **Primary Font:** Inter (Google Font)
- **Headings:** 700 weight, 2.25rem (h1) to 1.25rem (h4)
- **Body:** 400 weight, 1rem (medium), 0.875rem (small)

Responsive Design

Breakpoint	Device	Layout
< 640px	Mobile	Single column, compressed spacing
640-1024px	Tablet	Two columns, adjusted typography
> 1024px	Desktop	Multi-column, full spacing

Accessibility (WCAG 2.1 AA)

- Semantic HTML (h1-h6, main, nav, header, footer)
- Keyboard navigation (tab order, focus indicators)
- Screen reader support (ARIA labels, alt text)
- Color contrast minimum 4.5:1

Key UI Components

- **Button:** Primary, Secondary, Tertiary variants
- **Score Display:** Circular or bar with label and status
- **Skill Tag:** Name, category, level indicator
- **Upload:** Drag-and-drop or browse
- **Loading:** Spinner during processing
- **Card:** For section grouping
- **Accordion:** For expandable sections
- **Notification:** Success, error, warning messages

AI Integration

Google Gemini API

Version: Gemini 1.5 Pro

Prompt Structure

Resume Analysis Prompt:

text

You are an expert resume evaluator. Analyze the following resume and provide comprehensive evaluation with actionable insights.

RESUME CONTENT:

{extracted_text}

Provide:

1. ATS compatibility score (0-100) with breakdown
2. Professional summary
3. Strengths (list with brief explanation)
4. Weaknesses (list with brief explanation)
5. Missing skills (with priority 1-5)
6. Improvement suggestions (with priority 1-5)

7. Interview readiness score (0-100) with explanation
8. 10 interview questions (technical, behavioral, situational)

Format as JSON:

```
{
  "ats_score": number,
  "ats_breakdown": string,
  "summary": string,
  "strengths": [string],
  "weaknesses": [string],
  "missing_skills": [{ "skill": string, "priority": number }],
  "suggestions": [{ "suggestion": string, "priority": number }],
  "readiness_score": number,
  "readiness_explanation": string,
  "interview_questions": [{ "question": string, "type": string, "difficulty": string }]
}
```

ATS Scoring Logic

Factor	Weight	Description
Keyword Density	25%	Against industry standards
Formatting & Structure	20%	Section headers, consistency
Content Completeness	20%	All required sections present
Grammar & Spelling	10%	Accuracy and professionalism
Action Verb Usage	10%	Strong, varied verbs
Quantification	10%	Metrics and measurable achievements
Contact Information	5%	Complete and professional

Score Interpretation

Score	Status	Interpretation
90-100	Excellent	Well-optimized for ATS
75-89	Good	Room for improvement
60-74	Fair	Some issues affecting performance
0-59	Needs Improvement	Significant optimization needed

Interview Question Generation

Question Types:

- **Technical (40%):** Based on identified skills
- **Behavioral (30%):** Based on experience
- **Situational (20%):** Job-specific scenarios
- **General (10%):** Career goals and interests

Difficulty Calibration:

- **Entry-level:** Easy to Medium
- **Mid-level:** Medium to Hard
- **Senior-level:** Hard to Expert

Security

Security Principles

- **Defense in Depth:** Multiple layers of security controls
- **Least Privilege:** Minimum required permissions
- **Security by Default:** Secure configuration as default
- **Fail Secure:** Error conditions maintain security

File Upload Security

- **Type Validation:** MIME type and extension check
- **Size Limit:** 10MB maximum
- **Content Validation:** Scan for malicious content
- **Temporary Storage:** Delete after processing
- **Path Security:** Prevent directory traversal

Input Validation

Client-Side: Real-time validation, prevent invalid submissions

Server-Side: Re-validate all client-side checks, business rule validation

Validation Rules:

- Email format validation
- File type and size validation
- Text length validation (max 50,000 characters)
- Required field validation

SQL Injection Prevention

- **Parameterized Queries:** Always use prepared statements
- **Input Sanitization:** Escape special characters
- **Database Permissions:** Minimum required access
- **Type Validation:** Validate input data types

XSS Prevention

- **Content Sanitization:** Sanitize all user input
- **React Auto-Escaping:** React's built-in XSS protection
- **Security Headers:** X-XSS-Protection, Content-Security-Policy

API Security

- **Rate Limiting:** Prevent abuse
- **CORS Configuration:** Whitelist trusted origins
- **Input Validation:** Validate all request parameters
- **Error Handling:** No sensitive info in errors

Testing Strategy

Unit Tests (50%)

- **Frontend:** Component tests (Jest + React Testing Library)
- **Backend:** Service tests (Jest)
- **Utilities:** Function tests

Integration Tests (30%)

- API endpoint testing (Supertest)
- Database integration testing
- AI service integration testing
- File parsing tests

System Tests (15%)

- End-to-end testing (Cypress or Playwright)
- Workflow testing
- Performance testing
- Security testing

Acceptance Tests (5%)

- User story verification
- Requirements validation
- Business process testing

Edge Cases

Scenario	Expected Behavior
Empty PDF	Error: "Empty file"
Corrupted file	Error: "File corrupted"
File with no text	Error: "No text extracted"
Large file (10MB+)	Rejected with size limit message
Empty text input	Error: "No text entered"
Very long text	Rejected with length limit message

API timeout	Retry with error message
Database connection lost	Error: "Service unavailable"

Deployment Strategy

AWS EC2 Setup

```
bash
# Update system
sudo apt update && sudo apt upgrade -y

# Install Node.js 18
curl -fsSL https://deb.nodesource.com/setup_18.x | sudo -E bash -
sudo apt install -y nodejs

# Install PM2
sudo npm install -g pm2

# Deploy application
cd /var/www/smarthire
npm install
npm run build

# Start with PM2
pm2 start src/app.js --name smarthire-api
```

Security Group Rules

Type	Protocol	Port	Source	Purpose
SSH	TCP	22	Developer IP	SSH access
HTTP	TCP	80	0.0.0.0/0	Web access
HTTPS	TCP	443	0.0.0.0/0	Secure web access
Custom	TCP	3000	0.0.0.0/0	Application access

AWS RDS Setup

- Engine: MySQL 8.0
- Class: db.t2.micro
- Storage: 20GB
- Automated backups: 7 days retention
- Public Accessibility: No

Environment Variables

env

NODE_ENV=production

PORT=3000

DB_HOST=smarthire-db.xxxxxxxxxxxxxx.rds.amazonaws.com

DB_USER=admin

DB_PASSWORD=secure_password

DB_NAME=smarthire_ai

GEMINI_API_KEY=your_api_key

GEMINI_MODEL=gemini-1.5-pro

CORS_ORIGIN=https://smarthireai.com

MAX_FILE_SIZE=10485760

LOG_LEVEL=info

Future Roadmap

Version 1.1

Authentication Implementation:

- User registration and login
- Password management
- Session handling
- Role-based access control

Enhanced Features:

- Multiple resume management
- Resume version history
- Side-by-side comparison
- Email notifications

Version 2.0

Recruitment Integration:

- Job posting and management
- Candidate matching
- Automated screening
- Interview scheduling

Advanced Analytics:

- Trend analysis
- Predictive analytics
- Industry benchmarks

Integration Capabilities:

- LinkedIn integration
- Job board connections
- Webhook support

Version 3.0

Enterprise Features:

- Team collaboration
- Workflow automation
- Custom reporting
- Advanced security

Platform Expansion:

- Mobile applications
- API-first architecture
- Third-party ecosystem
- White-label solutions

AI Enhancement:

- Custom model training
- Multi-language support
- Industry-specific modules
- Continuous learning

Success Metrics

Technical Metrics

Metric	Target	Measurement
System Uptime	99.5%	Application monitoring
API Response Time	< 2s	Performance monitoring
AI Processing Time	< 30s	Custom metrics
Error Rate	< 5%	Error tracking
File Upload Success	> 95%	Upload tracking

User Metrics

Metric	Target	Measurement
Analysis Completion	> 80%	User flow tracking
User Engagement	> 50% return	Analytics
Recruiter Usage	> 50% active	Database tracking
Report Downloads	> 40%	Download tracking
Quality Metrics		
Metric	Target	Measurement
User Satisfaction	> 80% satisfied	Survey feedback
AI Accuracy	> 85%	Manual validation
Suggestion Usefulness	> 75% helpful	User feedback
Feature Adoption	> 60%	Analytics tracking

Conclusion

SmartHire AI is a comprehensive, production-ready recruitment platform demonstrating full-stack development, AI integration, cloud computing, database design, and responsive UI/UX. This document provides complete specifications for development, testing, and deployment.

The platform bridges the gap between job seekers and modern recruitment systems through AI-powered resume analysis, actionable improvement suggestions, and personalized interview preparation. The intuitive interface for candidates and recruiters ensures accessibility while the robust technical architecture guarantees performance, reliability, and future scalability.

All key software engineering concepts are demonstrated:

- Full-stack JavaScript development with React.js and Node.js
- REST API design and implementation
- Relational database design with MySQL
- AWS cloud computing (EC2, RDS)
- Google Gemini AI integration
- Responsive UI/UX with accessibility standards
- Comprehensive testing and security practices

This document enables immediate implementation by any engineering team with complete clarity on requirements, specifications, and quality standards.